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MULTIPLE CHOICE QUESTIONS (MCQ'S)

1. The value of elastic restoring forces in case of a spring is:
   a. Kx b. –Kx c. ¼ Kx d. -1/4 Kx

2. If the bob of a simple pendulum is replaced by another bob of double mass but of the same size, its time period:
   a. Increases b. Decreases **c. Remains the same** d. becomes infinity

3. The time period of a simple pendulum depends upon its:
   a. Length b. Amplitude c. Temperature d. Mass of the bob

4. An object is executing simple harmonic motion. Its kinetic energy is maximum at its:
   a. mean position **b. extreme position**
   c. at any point along the path d. None of these

5. A spring mass system is performing simple harmonic motion will the time period 'T'. If we double the mass of its bob, the new time period will be:
   a. T b. 2T c. 1.414T **d. 0.707T**

6. The acceleration of free fall on the moon is 1/6 of that on the earth. What would be the period on the moon of a simple pendulum which has a period of 1 second on the earth.
   a. 1/6 sec b. 1/6 sec **c. 6 sec** d. 12 sec

7. The frequency of oscillation of a simple pendulum depends upon:
   a. The mass of the bob b. The amplitude of vibration
   c. The light of the pendulum d. None of them

8. What is the defining of oscillation of a simple pendulum depends upon:
   a. The mass of the bob b. The amplitude of vibration
   c. The light of the pendulum d. None of them

9. What is the defining characteristic of damped oscillations?
   a. Constant amplitude b. Increasing amplitude
   c. Decreasing amplitude d. No amplitude

10. In a damped oscillatory system, the damping force is proportional to:
    a. Constant amplitude b. Increasing amplitude
    c. Decreasing amplitude d. No amplitude

11. In a damped oscillatory system, the damping force is proportional to:
    a. Displacement b. Velocity c. Acceleration d. Mass

12. What happens to the amplitude of a free oscillation over time?
    a. Displacement **b. Velocity** c. Acceleration d. Mass

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13. What happens to the amplitude of a free oscillation over time?
    a. It remains constant   b. It decreases   c. It increases   d. It varies unpredictably

14. In a damped oscillatory system, the damping force is proportional to:
    a. Displacement   b. Velocity   c. Acceleration   d. Mass

15. What happens to the amplitude of a free oscillation over time?
    a. It remains constant   b. It decreases   c. It increases   d. It varies unpredictably

16. Which of the following is a common source of damping in mechanical oscillatory systems?
    a. Friction   b. Gravity   c. Tension   d. Elasticity

17. In order to double the period of a simple pendulum:
    a. Its length should be doubled   **b. Its length should be quadrupled**
    c. Its mass should be double   d. Its mass should be quadrupled

18. A simple Harmonic oscillator has amplitude A and time period t. Its maximum speed is:
    a. 4 A/t   b. 2A/t   c. $4\pi A/t$ **d. $2\pi A/t$**

19. A spring attached by a load of weight W is vibrating with a period T. If the spring is divided in four equal parts and the same load is suspended from one of these parts, the new time period is:
    a. T/4   b. 2T   **c. T/2**   d. 4T

20. The total energy of a particle executing simple harmonic motion is proportional to:
    a. frequency of oscillation   b. maximum velocity of motion
    c. amplitude of motion   **d. square of amplitude of motion**

21. A child swinging on a swing in sitting position, stands up, then the time period of the swing will:
    a. Increase   **b. Decrease**   c. Remains the same
    d. Increases if the child is long and decreases if the child is short

22. If a body oscillates at the angular frequency cod of the driving force, then the oscillations are called:
    a. Forced oscillations   b. Coupled oscillations   c. Free oscillations d. Maintained oscillations

23. A simple harmonic oscillator with a natural frequency ω d. The Resonance occurred when:
    a. ω n < ω d   b. ω n > ω d   **c. ω n = ω d**   d. ω n ≠ ω d

24. In vehicles, shock absorber reduced the jerks:
    a. The shock absorber is the application of damped oscillations
    b. Damping effect is due to the fractional loss of energy
    c. Shock absorbers in vehicles reduced jerk
    d. All of these

25. A heavily damped system has a fairly flat resonance curve in:
    a. An acceleration time graph
    b. An amplitude frequency graph
    c. Velocity time graph
    d. Distance-time graph

26. If the mass of the bob of a simple pendulum is doubled, its time period will:
    a. doubled
    b. Halved
    c. Remain constant
    d. Quadrupled

27. What is the mathematical representation of Kirchhoff's Voltage Law (KVL) for a closed loop in an electrical circuit?
    a. $\sum V = 0$
    b. $\sum I = 0$
    c. V = IR
    d. P = IV

28. When a resistor carries a current (i) of 2 amperes and has a resistance (R) of 4 ohms, what is the power dissipated in the resistor?
    a. Conversion of heat into mechanical energy
    b. Conversion of heat into electrical energy
    c. Conversion of mechanical energy into heat
    d. Conversion of electrical energy into heat

29. If two resistance of 4 ohm and 8 ohm are connected in series and 2 Ampere current is flowing through 4 ohm resistance what will be current through 8 ohms resistance.
    a. 4 amp
    b. 8 amp
    c. 2 amp
    d. 1 amp

30. In a parallel-plate capacitor, the electric field is:
    a. Uniform
    b. Non-uniform
    c. Zero
    d. Random

31. What is the primary purpose of a Wheatstone bridge circuit?
    a. To measure voltage
    b. To measure current
    c. To measure resistance
    d. To generate electrical power

32. In a potentiometer, what is the principle of measurement used?
    a. Ohm's Law
    b. Faraday's Law
    c. Joule's Law
    d. None of these

33. What is the purpose of color coding on a resistor?
    a. To indicate its physical size
    b. To specify its operating temperature range
    c. To show its only tolerance value
    d. To denote its resistance value

34. If a resistor has the color bands brown, black, and red, what is its resistance value?
    a. 100 ohms
    b. 1000 ohms
    c. 10,000 ohms
    d. 1,000,000 ohms

35. What is the SI unit of resistivity?
    a. Ohm (Ω)
    b. Ohm-meters (Ω m)
    c. Amperes (A)
    d. Volta (V)

36. If you double the length of a wire while keeping its cross-sectional area constant, how does this affect its resistance?
    a. Resistance doubles
    b. Resistance quadruples
    c. Resistance remains the same
    d. Resistance halves

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37. How does the resistivity of most conductors change with increasing temperature?
    a. It decreases
    b. It remains constant
    c. It increases
    d. It depends only on the specific material

38. Which unit is used to measure conductance?
    a. Ohms (Ω)
    b. Siemens (S)
    c. Amperes (A)
    d. Volts (V)

39. What is the source of EMF in a battery?
    a. Chemical reactions
    b. Magnetic fields
    c. Friction
    d. Gravitational force

40. In an ideal EMF source, what is the internal resistance?
    a. Zero
    b. Infinity
    c. Variable
    d. Equal to the external resistance

41. In an ideal EMF source with no internal resistance, how does the terminal voltage compare to the EMF when a load is connected?
    a. Terminal voltage is equal to EMF
    b. Terminal voltage is greater than EMF
    c. Terminal voltage is less than EMF
    d. Terminal voltage is zero

42. Kirchhoff's laws are useful in determining:
    a. current flowing in a circuit
    b. e.m.f. and voltage drops in it a circuit
    c. power in a circuit
    d. only e.m.f. in a circuit

43. The resistance of a superconductor is:
    a. finite
    b. infinite
    c. changes with every conductor
    d. zero

44. Reciprocal of resistance is called:
    a. conductance
    b. resistivity
    c. resonance
    d. capacitance

45. The graphical representation of Ohms
    a. parabola
    b. hyperbola
    c. ellipse
    d. straight line

46. A potential difference is applied across the ends of a wire. If the potential difference is doubled, then the drift velocity of free electrons will:
    a. be quadrupled
    b. be doubled
    c. be halved
    d. remain unchanged

47. Internal resistance is the resistance offered by:
    a. Capacitor
    b. Resistor
    c. Conductor
    d. Source of emf

48. Power dissipation in a resistor can be calculated using which formula:
    a. $P = V^2/R$
    b. $P = I^2 \times R$
    c. $P = V \times I$
    d. $P = R / (V \times I)$

49. What is a potentiometer primarily used for?
    a. Measuring electric current
    b. Measuring electric charge
    c. Measuring potential difference (voltage)
    d. Measuring electric resistance

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50. A heat-sensitive device whose resistivity changes with the change in temperature is called:
    a. conductor   b. resistor   **c. thermistor**   d. thermometer

51. A wire of uniform area of cross-section A length L and resistance R is cut into two parts. The resistivity of each part:
    a. Becomes zero   b. is halved   c. is doubled   **d. remains same**

52. According to Ohm's Law, if the voltage across a resistor is doubled, what happens to the current through it?
    a. It doubles   b. It halves   c. It remains the same   d. It quadruples

53. During the discharging of a capacitor, what happens to the voltage across the capacitor over time?
    a. It increases exponentially   **b. It decreases exponentially**
    c. It remains constant   d. It oscillates

54. In an RC circuit, if the resistance (R) is increased while the capacitance (C) is kempt constant, what happens to the time constant (i)?
    a. It decreases   **b. It increases**   c. It remains the same   d. It becomes zero

55. What is the primary function of a parallel plate capacitor?
    a. To store energy in the form of magnetic fields   **b. To store energy in the form of electric fields**
    c. To regulate current flow in a circuit   d. To amplify electrical signals

56. In a parallel plate capacitor, if you increase the distance between the plates while keeping the charge and plate area constant, what happens to the capacitance?
    a. It increases   **b. It decreases**   c. It remains the same   d. It becomes zero

57. If the voltage across a parallel plate capacitor is doubled while keeping the charge constant, what happens to the electric field strength between the plates?
    a. It doubles   b. It halves   c. It remains the same   d. It becomes zero

58. When a dielectric material is inserted between the plates of a parallel plate capacitor, what effect does it have on the capacitance?
    a. Increases the capacitance   b. Decreases the capacitance
    c. Has no effect on capacitance   d. Reverses the polarity of the plates

59. When capacitors are connected in series, what happens to their total capacitance compared to individual capacitances?
    a. It increases   b. It decreases   c. It remains the same   d. It becomes zero

60. The charging of a capacitor through a resistance follows:
    a. Linear law   b. Square law   **c. Exponential law**   d. None of the above

61. When the total charge in a capacitor is doubled, the energy stored:
    a. remains the same   b. is doubled   c. exponential law   **d. None of these**

62. When the total charge in a capacitor is doubled, the energy stored:
    a. remains the same   b. is doubled   c. is halved   d. is quadrupled

63. The capacitance C is charged through a resistor R. The time constant of the charging circuit is given by:
    a. C/R   b. 1/RC   c. RC   d. R/C

64. Capacitor blocks:
    a. alternating current   **b. direct current**
    c. both alternating and direct current   d. neither alternating nor direct current

65. What type of dielectric material is commonly used in capacitors to increase their capacitance?
    a. Non-conductive   b. Conductive   c. both alternating and direct current
    d. neither alternating nor direct current

66. What type of dielectric material is commonly used in capacitors to increase their capacitance?
    a. Non-conductive   **b. Conductive**   c. Magnetic   d. Metallic

67. How many conductors does the capacitor consist of?
    a. One   b. Two   c. Three   **d. Four**

68. What is the standard unit of electric charge?
    a. Watts   b. Henry   c. Farads   **d. Coulombs**

69. The time constant (T) in an RC circuit determines:
    a. The energy stored in the capacitor   b. The rate of charging of the capacitor
    c. The resistance of the circuit   d. The capacitance of the capacitor

70. If you double the capacitance (C) in an RC circuit while keeping the resistance (R) constant, what happens to the time constant (T)?
    a. It doubles   b. It halves   c. It remains unchanged   d. It becomes zero

71. The capacitance of a capacitor is NOT influenced by:
    a. Plate thickness   b. Plate area   c. Plate separation   d. Nature of the dielectric

72. What is the value of capacitance of a capacitor which has a voltage of 4V and has 16C of charge?
    a. 2F   **b. 4F**   c. 6F   d. 8F

73. Capacitors are used in electric power supply system to:
    a. Improve resistance   b. Reduce line current   **c. Provide voltage stability** d. switching

74. In a variable hi a variable capacitor, capacitance can be varied by:
    a. Turning the rotatable plates in or out   b. Changing the plates
    c. Sliding the rotatable plates   d. Changing the material of plates

75. Energy stored in the capacitor is:
    a. E = 1/4CV   **b. $E = 1/2CV^2$**   c. $E = CV^2$ d. E = 1/2CV

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76. The time constant of a series RC circuit consisting of 100J.F capacitor in series with a 100Ω resistor is:
    a. 0.1s b. 0.1ms **c. 0.01s** d. 0.01ms

77. The electric flux through a surface will be minimum, when the angle between E and ΔA is:
    a. 90^o b. zero c. 45^o d. 60^o

78. Which of the following cannot be the unit of electric intensity?
    a. N/C b. V/m c. J/C-m **d. J/G**

79. Electron volt is the unit of:
    a. Energy b. Force c. Potential difference d. Current

80. One joule per coulomb is equal to:
    a. Ampere **b. Volt** c. Farad d. Tests

81. The force per unit charge is known as:
    a. Electric flux **b. Electric field intensity** c. Electric potential d. Electric current

82. The Electric flux through a closed surface depends upon the:
    a. Magnitude of the charge enclosed by the surface b. Position of the charge enclosed by the surface
    c. The shape of the surface d. None of the above options

83. In an electric field, the negative space rate of variation of potential at a point is equal to __________at a point.
    a. Potential gradient b. Absolute potential c. Electric intensity d. Electric potential

84. Decreasing the separation of two positive charges by one-half will cause the force of repulsion to change by:
    a. 4 times b. 2 times c. ½ times d. ¼ times

85. In Radio and Television broadcast, the information signal is in the form of:
    a. analog signal b. digital signal
    c. both analog and digital signals d. neither analog nor digital signal

86. A communication channel consists of:
    a. transmission line only b. optical fiber only c. free space only d. all of them

87. Voltage signal generated by a microphone is:
    a. digital in nature **b. analog in nature** c. hybrid nature d. consists of bits & bytes

88. The number of zones in n=2 bit quantization is:
    a. 2 **b. 4** c. 8 d. 16

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89. What is the modulation index of an amplitude-modulated (AM) signal with a maximum amplitude of 10V and a minimum amplitude of 2V?
    a. 5 b. 4 c. 3 d. 2

90. The sequence of conversion of Analog signal to digital signal in A/D conversion is:
    a. Quantization. Sampling and Encoding b. Encoding, sampling and quantization
    c. Sampling, encoding and quantization **d. Sampling, Quantization and encoding**

91. We can convert sound waves into electrical oscillations by _____.
    a. Microphone b. Television c. Radio d. None of these

92. On which component modulation is performed:
    a. Transmitter b. Signal c. Jammer d. None

93. Which of the following is the core of an optical fiber?
    a. Plastic **b. Glass** c. Copper d. Air

94. When a beam of while light passes through a prism, it splits in to different colors. The phenomenon is called:
    a. Dispersion b. Polarization c. Diffraction d. Refraction

95. Light of wavelength λ is incident normally on a diffraction grating having grating element (d). The angle between the zero order and first order spectra depends upon:
    a. λ only b. d only c. Both d and λ d. None

96. The central fringe in Young's double slit experiment is always:
    a. Bright b. Dark c. Dark or Bright both are possible d. Red

97. Diffraction of x-ray can be studied by using:
    a. A thin film b. Diffraction greting c. Rock salt d. Tourmaline crystal

98. If a number of waves of a beam vibrate in number of direction, then the beam of light is called:
    a. Polarized b. Unpolarized c. Plane polarized d. Circular polarized

99. When a movable mirror in the Michelson's interferometer is moved by $\lambda/4$ the path difference:
    a. Bright b. Dark c. Dark or bright both are possible d. Red

100. When a movable mirror in the Michelson's interferometer is moved by $\lambda/4$ the path difference:
    a. $\lambda/4$ b. $\lambda/2$ c. 2λ d. λ

101. Electromagnetic waves consist of electric and magnetic field. Both fields are:
    a. Parallel to each other b. Parallel to the direction of propagation
    c. Perpendicular to each other d. Perpendicular to the direction of propagation

102. The characteristic property of light, which does not change with the medium, is:
    a. Wave length b. Frequency c. Velocity d. All of these

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103. If λ is the wave length of light in air then the wave length in a denser medium is:
a. λ b. $n\lambda$ c. $\lambda/2n$ d. λ/n

104. The structure of x-rays of a crystal can be studied with the help of ________of x-rays.
a. Interference b. Diffraction c. Polarization d. Absorption

105. Diffraction of light is a special type of:
a. Reflection b. Polarization c. Refraction d. Interference

106. In Michelson interferometer, semi-silvered plate is used to obtain:
a. Dispersion b. Phase coherence c. Monochromatic light d. Unpolarized light

107. Which of the following are not electromagnetic rays?
a. X-ray b. Radio waves c. Ultra violet rays d. α-rays

108. Which of the following communication channels is the most suitable for long distance communication?
a. Optical fiber b. Radio waves c. Twisted pair cable d. Human voice

109. What are the advantages of digital circuits:
a. Less noise b. Less interference c. More flexible **d. All of the above**

110. Phase Modulation is a type of:
a. Angle Modulation b. Amplitude Modulation c. Digital Modulation d. Pulse Modulation

111. Which modulation technique is more immune to amplitude noise and interference?
a. AM (Amplitude Modulation) **b. FM (Frequency Modulation)**
c. PM (Phase Modulation) d. ASK (Amplitude-Shift Keying)

112. The frequency range (20 – 20,000 Hz) is called _______range:
a. Audible b. Electromagnet c. Simple d. None of these

113. Advantages of Digital communication is:
a. Easy to convert **b. Multiplexing** c. Used for short distance d. Difficult to store

114. In frequency modulation carrier signal always have ______frequency than modulating frequency:
a. Low **b. High** c. Same d. Zero

115. In AM modulation, Modulation index will always be less then or equal to:
a.1 b. 10 c. 0 d. -1

116. Electromagnetic waves from an unknown source in space are found to be diffracted when passing through gaps of the order of 10^{-4}m, which type of the wave are they most likely to be?
a. microwaves b. Ultraviolet c. Radio waves d. Infra-red waves

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117. Huygens's conception of secondary waves:
    a. Helps us to find the focal length of a thick lens
    b. is a geometrical method to find a wave front
    c. Is used to determine they velocity of light
    d. is used to explain the polarization of light

118. Interference fringes are produced using monochromatic light of same intensity from a double slit screen. If the intensity of light emerging from one of the slit is reduced, the effect on interference pattern will be:
    a. All the dark and bright fringes become brighter
    b. All the dark and bright fringes become darker
    c. Bright fringes become brighter and dark fringes become darker
    d. Bright fringes become darker and dark fringes become brighter

119. In Young's experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe width will be:
    a. 4 times b. ¼ times c. doubled d. unchanged

120. A ray of light passes from air into water. Striking the surface of the water with an angle of incidence 45^{o}. Which of these quantities change as the light enters the water. i) wavelength ii) frequency iii) speed of propagation iv) direction of propagation:
    a. I and II only b. III and IV only c. I, III, and IV only d. all of them

121. A hill separates a television (TV) transmitter from a house. The transmitter cannot be seen from the house but still the TV in the house has good reception. What wave phenomena make it possible?
    a. Coherence of waves **b. Diffraction of waves** c. Interference of waves d. Refraction of waves

122. Monochromatic light is incident on a diffraction grating and a diffraction pattern is observed. Which effects is observed by replacing the grating with one that has more lines per millimeter?
    a. Number of maxima decreases with decrease in angle between first and second order maxima
    b. Number of maxima decreases with increase in angle between first and second order maxima
    c. Number of maxima increases with decrease in angle between first and second order maxima
    d. Number of maxima increases with increase in angle between first and second order maxima

123. Optically active substances are those substances which:
    a. Produce polarized light b. Rotate the plane of polarization of polarized light
    c. Produce double refraction
    d. Convert a plane polarized light into circularly polarized light

124. If the wavelength of an electromagnetic wave is about the diameter of a cricket ball, what type of radiation is it.
    a. X-ray b. Ultraviolet c. Radio waves d. Visible light

125. As compared to sound waves frequency of radio waves is:
    a. higher b. equal c. lower d. may be higher or lower

126. The process of superimposing signal frequency on the carrier wave is known as:
a. transmission b. detection c. reception **d. modulation**

127. What is the frequency range of signals that can be transmitted in case of twisted pair of wires?
a. 10MHz to 15MHZ b. 5MHz to 10MHz **c. 100Hz to 5MHz** d. 10Hz to 100Hz

128. The maintenance of a satellite's orbital position is called:
a. maintenance keeping **b. station keeping** c. station maintenance d. attitude control

129. Process of mapping the sampled analog voltage values to discrete voltage levels is called:
a. sampling b. sampling frequency c. quantizing d. encoding

130. AM is used for broadcasting because:
a. it requires less transmitting power compare with other systems
b. It is more noise immune than other modulation system
c. No other modulation can provide the necessary bandwidth faithful transmission
d. Its use avoids receiver completely

131. Data in compact disc is stored in the form of:
a. analog signal b. digital signal c. noise d. both a and b

132. Which of the following is not a communication channel?
a. Optical fiber b. Radio waves c. Twisted pair cable **d. Human voice**

133. Twisted Pair Cable are suitable for:
a. Long Range b. Short Range c. Medium Range **d. Both b and c**

134. In a Digital-to-Analog Converter (DAC), what is the process of converting digital values into a continuous analog signal called?
a. Demodulation b. Sampling c. Modulation **d. Reconstruction**

135. Plane polarized light is passed through a Polaroid. On viewing through the polaroid we find that when polaroid is given one complete rotation about the direction of light:
a. The intensity of light gradually decreases to zero and remains at zero
b. The intensity of light gradually increases to maximum and remains at maximum
c. There is no change in the intensity of light
d. The intensity of light varies such that it is twice maximum and twice zero

136. Which of the following is the highest frequency region of the electromagnetic spectrum?
a. Radio waves b. Microwaves c. Visible light **d. Gamma**

137. Which of the following electromagnetic waves the longest wavelength?
a. Radio waves b. Microwaves c. Visible light d. Gamma rays

138. Which of the following electromagnetic waves is used in microwave ovens?
a. Radio waves **b. Microwaves** c. Visible light d. Gamma rays

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139. Which of the following electromagnetic waves is used in medical x-ray machines?
a. X-rays b. Gamma rays c. Ultraviolet radiation d. Microwaves

140. Which of the following electromagnetic waves is used in remote controls for TVs and other devices?
a. Infrared radiation b. Visible light c. Ultraviolet radiation d. Microwaves

141. Which one of the following proves that light is transverse in nature.
a. Interference b. Polarization c. Diffraction d. Refraction

142. Experimental demonstration of wave nature of light was provided by:
a. Thomas Young b. Compton c. Huygen d. Maxwell

143. Electromagnetic waves theory was given by:
a. Newton b. Compton c. Huygen **d. Maxwell**

144. Photoelectric effect can be explained if the light is considered to travel in the form of:
a. Waves **b. Particles** c. Both waves and particles d. Electron

145. A point source of light placed in a homogenous medium produces a:
a. Plane wave front **b. Spherical wave front** c. Cylindrical wave front d. Circular wave front

146. In Young's double slit experiment the fringes width or fringe spacing is equal to:
a. λ d/L b. λ dL c. 2λ d/L d. λ L/d

147. If the pressure of a gas is reduced to one-fourth of its initial pressure, what happens to the speed of sound in the gas, assuming other factors remain constant?
a. It increases by a factor of 2 b. It decreases by a factor of 2
c. It increases by a factor of $\sqrt{2}$ **d. Remain same**

148. In the context of sound waves, what does 'superposition' refer to?
a. The combination of two or more waves to form a new wave
b. The reflection of sound waves off a surface
c. The absorption of sound energy by a medium
d. The scattering of sound waves in all directions

149. If one tuning fork vibrates of 440 Hz and another at 443 Hz, what is the beat frequency produced when they are sounded together?
a. 440Hz **b. 3Hz** c. 443Hz d. 880Hz

150. Seats are the result of the interference of sound waves with:
a. Identical frequencies b. Frequencies that are exactly out of phase
c. Slightly different frequencies d. Frequencies in different mediums

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151. What happens to the frequency of vibration of a stretched string when its tension is increased while keeping other factors constant?
a. The frequency decreases
b. The frequency remains the same
c. The frequency increases
d. The frequency becomes zero

152. Which of the following factors affects the frequency of vibration in a stretched string?
a. Temperature
b. Material of the string
c. Length of the string
d. All of the above

153. If you have a guitar string with a length of 60cm and you pluck it, producing a fundamental frequency of 440 Hz, what will be the fundamental frequency if you shorten the string to 30cm, keeping the tension constant?
a. 220 Hz
b. 440 Hz
c. 880 Hz
d. 660 Hz

154. If a wire has a fundamental frequency of 440 Hz when vibrating at room temperature, and you heat the wire, increasing its linear density, what will happen to the fundamental frequency?
a. it will decreases
b. it will remain the same
c. it will increases
d. it will depend on the wire's length

155. The condition of interference in thin films is reversed because of:
a. Small thickness of film
b. Refraction
c. Phase reversal
d. Diffraction

156. Which of the following represents Gregg's law?
a. $m\lambda = dn/0$
b. $2m\lambda = dsin0$
c. $m\lambda/2 = dsin0$
d. $m\lambda = dsin0/2$

157. Light posses:
a. Transverse nature
b. Electromagnetic
c. Dual nature
d. All of these

158. In thin films, destructive interference occurs when the path difference is:
a. An odd multiple of half wave length
b. An odd multiple of wave length
c. An integral multiple of half wave length
d. An integral multiple of wave length

159. The number of lines ruled per centimeter is on a diffraction grating is 4000. Its grating element is:
a. 2.5×10^{-6} m
b. 2.5×10^{-4} m
c. 2.5×10^{-7}m
d. 2.5×10^{-8} m

160. The bending of light around sharp corners of an obstacle is called:
a. Diffraction
b. Interference
c. Polarization
d. Refraction

161. The dispersion of white light after passing through a prism is due to:
a. Different intensities
b. Different amplitude
c. Different temperature
d. Different wave lengths

162. Light year is a unit of:
a. Energy
b. Time
c. Distance
d. Intensity

163. The condition for constructive and the destructive interference are reversed in case of thin film due to:
a. Phase reversal of one side of the wave
b. Phase reversal of both parts of the wave
c. Phase reversal of the none
d. Change in frequency of the wave

164. To replace a bright fringe by the next bright fringe in a Michelson Interferometer, the moveable mirror is moved through a distance equal to:
a. $\lambda/4$ **b. $\lambda/2$** c. 2λ d. λ

165. Which of the following is TRUE about the single slit diffraction pattern?
a. The central maximum is the brightest b. The secondary maxima are equally spaced
c. The width of the central maximum is inversely proportional to the wavelength of light
d. All of the above are TRUE

166. In a damped forced oscillation, what is the phase relationship between the driving force and the displacement of the system at resonance?
a. 0 degrees (in-phase) **b. 90 degrees (out of phase)**
c. 180 degrees (opposite phase) d. It varies with damping

167. In a simple harmonic oscillator, when the displacement from the mean position is maximum, which energy is maximum?
a. Kinetic energy b. Potential energy c. They are equal d. It depends on the specific oscillator

168. At the mean position of a simple harmonic oscillator, the kinetic energy is:
a. Zero **b. Maximum**
c. Equal to potential energy d. Half of the potential energy

169. In an oscillatory system, which type of energy is associated with the displacement of the object from its equilibrium position?
a. Kinetic energy **b. Potential energy**
c. Both kinetic and potential energy d. None of the above

170. The total mechanical energy of an oscillator in simple harmonic motion.
a. Increases with time b. Decreases with time
c. Remains constant d. Depends on the mass of the oscillator

171. In a simple harmonic oscillator, which energy is at its maximum when the displacement is zero?
a. Kinetic energy b. Potential energy c. Both are zero d. They are always equal

172. A particle is executing SHM with amplitude A and angular frequency ω. What is the phase angle of the particle at a time t = T/4?
a. 0^o b. 90^o c. 180^o d. 270^o

173. The speed of sound in air at STP is 332 m/s. If the air pressure becomes double at the same temperature, the speed of sound become:
a. 1382 m/s b. 664 m/s **c. 332 m/s** d. 166 m/s

174. How does the speed of sound v in air depend on the atmospheric pressure P:
a. $v \propto P^o$ b. $v \propto P^{-1}$ c. $v \propto P^2$ d. $v \propto P^1$

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175. The speed of sound in a gas is proportional to:
a. square root of isothermal elasticity
b. square root of adiabatic elasticity
c. isothermal elasticity
d. adiabatic elasticity

176. The length of a pipe closed at one end is L. In the standing wave whose frequency is 7 times the fundamental frequency, what is the closest distance between nodes?
a. $\frac{1}{L4}L$
b. $\frac{1}{7}L$
c. $\frac{2}{7}L$
d. $\frac{4}{7}L$

177. A 620 Hz frequency song of a ice cream trolley approaches with speed to a boy standing at the door of his house is heard with frequency f_1. If the trolley is stopped and the boy approaches to the ice cream trolley with same speed v_1 the boy now hears the sound with frequency f_2. Choose the correct statement:
a. $f_1 = f_2$; both are greater than 620 Hz
b. $f_2 > f_1 > 620$ Hz
c. $f_1 = f_2$; both are lesser than 620 Hz
d. $f_1 > f_2 > 620$ Hz

178. The speed of sound in a gas in which two waves of wavelength 50cm and 50.4 cm produce 6 beats per second is:
a. 338 m/s
b. 350 m/s
c. 378 m/s
d. 400 m/s

179. The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point in the medium in 2 minutes, then its wave length:
a. 13.85m
b. 25.3m
c. 41.5m
d. 57.2m

180. In Newton's formula, what does "y" represent?
a. Gravitational constant
b. Speed of sound
c. Adiabatic index (ratio of specific heats)
d. Density of the medium

181. Which parameter does Laplace's correction primarily modify in the speed of sound formula?
a. Frequency
b. Amplitude
c. Temperature
d. Pressure

182. What is the relationship between the speed of sound and temperature in a gas?
a. Directly proportional
b. Inversely proportional
c. No relationship
d. Randomly proportional

183. If the temperature of a gas is doubled while keeping other factors constant, what happens to the speed of sound in the gas?
a. It doubles
b. It becomes half
c. It remains the same
d. It becomes four times

184. When comparing open and closed pipes of the same length, which one produces a higher fundamental frequency?
a. Open pipe
b. Closed pipe
c. Both produce the same fundamental frequency
d. It depends on the material of the pipe

185. What is the Doppler effect?
  a. The change I the speed of light due to motion
  b. The change in the pitch of a sound due to motion
  c. The change in the amplitude of a wave due to motion
  d. The change in the wavelength of a wave due to motion

186. If an observer moves towards the stationary source of sound with the speed of sound, its apparent frequency is:
  a. Half the original frequency
  b. Double the original frequency
  c. Infinite
  d. Zero

187. Which type of wave experiences the Doppler effect?
  a. Only sound waves
  b. Only light waves
  c. All types of waves
  d. Only electromagnetic waves

188. A medical ultrasound machine is used to measure the speed of blood flow in a vein. The ultrasound machine emits a wave of frequency 5 MHz. The reflected wave has a frequency of 5.1 MHz. What is the speed of the blood flow? (Speed of sound in blood = 1540 m/s)
  a. 154 m/s b. 308 m/s c. 462 m/s d. 616 m/s

189. The speed v of a wave represented by $y = A\sin(\omega t - kx)$ is:
  a. k/ω **b. ω/k** c. ωk d. $1/\omega k$

190. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^3 are:
  a. 5, 1, 2 b. 5, 1, 5 c. 5, 5, 2 d. 4, 4, 2

191. Which among the following is the supplementary unit:
  a. Mass b. Time **c. Solid angle** d. Luminosity

192. The unit of solid angle is:
  a. second **b. Steradian** c. Kilogram d. Candela

193. The quantity having the same unit in all system of unit is:
  a. Mass **b. Time** c. Length d. Temperature

194. Random errors can be eliminated by:
  a. taking number of observations and their mean
  b. measuring the quantity with more than one instrument
  c. eliminate the cause
  d. careful observations

195. Systemic error can be:
  a. either positive or negative
  b. positive only
  c. negative only
  d. zero error

196. $[MLT^{-2}]$ is dimensional formula of:
a. strain **b. force** c. displacement d. pressure

197. Which of the following pair has the same dimension?
a. moment of inertia and torque **b. impulse and momentum**
c. surface tension and force d. specific heat and latent heat

198. Dependent variable is:
a. cause b. cause and effect **c. effect** d. reason

199. Dimension of kinetic energy is same as that of:
a. Acceleration **b. work** c. velocity d. force

200. What is the slope of the line represented by the equation y = 2x + 1?
a.2 b. 1 c. 0 d. Infinity

201. Which of the following is the definition of accuracy?
a. The closeness of a measurement to the true value
b. The closeness of repeated measurements to each other
c. Both a and b d. none of the above

202. Which of the following is the definition of precision?
a. The closeness of a measurement to the true value
b. The closeness of repeated measurements to each other
c. Both a and b d. None of the above

203. Which of the following is an example of a high-precision instrument?
a. A ruler b. A stopwatch **c. A balance** d. All of the above

204. 1 Which of the following is an example of a high-accuracy instrument?
a. A ruler b. A stopwatch **c. A balance** d. All of the above

205. What is resolution?
a. The smallest change in a quantity that can be measured or detected
b. The ability of an instrument to distinguish between two closely spaced objects
c. both a and b d. None of the above

206. Why is resolution important?
a. It allows us to make more accurate measurements
b. It allows us to see smaller details
c. Both a and b
d. None of the above

207. Which of the following has a higher resolution?
a. A digital camera with 10 megapixels or a digital camera with 20 megapixels?
b. A microscope with a magnification of 100x or a microscope with a magnification of 1000x?
c. A ruler with a scale of 1 cm or a ruler with a scale of 1 mm?
d. All of the above

208. Which of the following is not a benefit of high resolution?
a. Improved accuracy of measurements
b. Ability to see smaller details
c. Reduced noise in images and videos
d. Increased file size

209. What is the significance of resolution in the field of medicine?
a. High-resolution medical imaging devices allow doctors to diagnose diseases more accurately
b. High-resolution medical imagining devices allow doctors to perform surgery more precisely
c. Both a and b
d. None of the above

210. How many significant figures does the number 12.345 have?
a.1
b. 2
c. 3
d. 4

211. How many significant figures does the number 0.00345 have?
a. 1
b. 2
c. 3
d. 4

212. What is the correct answer to the following calculation, rounded to the correct number of significant figures? 123.4 + 58.78 = 180.18
a. 180
b. 180.1
c. 180.18
d. 180.2

213. Which of the following numbers has the same number of significant figures as the number 1.2345?
a. 123.45
b. 0.12345
c. 12.3450
d. 1.234500

214. Which of the following is true about significant figures?
a. Significant figures are the digits in a number that are reliable and meaningful
b. Significant figures include all non-zero digits and any zeros between them
c. Significant figures include all non-zero digits and any trailing zeroes
d. All of the above

215. What is extrapolation?
a. The process of estimating a value outside the range of the known data
b. The process of estimating a value inside the range of the known data
c. The process of finding the best fit line through a set of data points
d. None of the above

216. Which of the following is a derived quantity?
a. Length
b. Mass
c. Time
d. Velocity

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217. What is systematic error?
a. An error that is consistent in one direction, either positive or negative
b. An error that is unpredictable and varies randomly
c. An error that is caused by human error
d. An error that is caused by a faulty instrument

218. Which of the following is an example of systematic error?
a. A ruler that is too short
b. A balance that is not level
c. A stopwatch that is inaccurate
d. All of the above

219. Which of the following is an example of random error?
a. Reading the temperature from a thermometer at a different angle each time
b. Measuring the mass of an object on a balance that is not level
c. Using a stopwatch that is inaccurate
d. All of the above

220. The rate of change of linear momentum of a body is called a:
a. Linear force
b. Angular force
c. Power
d. Impulse

221. The term mass refers to the same physical concept as:
a. weight
b. Inertia
c. force
d. Acceleration

222. Which one of the following force is also called as self-adjusting force?
a. frictional force
b. tension
c. weight
d. thrust

223. The laws of motion shows the relationship between:
a. velocity and acceleration
b. mass and velocity
c. mass and acceleration
d. force and acceleration

224. Impulse is equal to:
a. Force x time
b. Momentum x time
c. Change in acceleration
d. Change in velocity

225. The modern view of friction takes into account the:
a. Roughness of the surfaces in contact
b. Adhesion between the surfaces in contact
c. Temperature of the surfaces in contact
d. All of the above

226. Which of the following is not a characteristic of static friction?
a. It is a self-adjusting force
b. It is proportional to the normal force
c. It is always in the direction opposite to the motion of the object
d. It can be overcome by applying a sufficiently large force

227. Which of the following is not an example of Newton's first law of motion?
a. A car moving at a constant speed on a straight road
b. A ball rolling down a hill
c. A book sitting on a table
d. A rocket taking off from the launch pad

228. One radian is about:
a. 25^o b. 45^o c. 37^o **d. 57^o**

229. Wheel turns with constant angular speed then:
a. each point on its rim moves with constant velocity
b. each point on its rim moves with constant acceleration
c. the wheel turns through equal angles in equal times
d. the angle through which the wheel turns in each second increases as time goes on

230. Which of the following factors will decrease the capillary rise of a liquid in a glass tube?
a. Increasing the diameter of the tube
b. Increasing the surface tension of the liquid
c. Decreasing the angle of contact between the liquid and the tube
d. Decreasing the density of the liquid

231. A hydraulic system has a piston with an area of 10 cm^3 and a piston with an area of $100cm^2$. If a force of 100N is applied to the smaller piston, what force will be produced by the larger piston?
a. 1000N b. 100N c. 10N d. 1N

232. Which of the following is the best explanation for why the drop of olive oil is perfectly spherical?
a. The olive oil is denser than the mixture of alcohol and water
b. The olive oil is less dense than the mixture of alcohol and water
c. The surface tension of the olive oil is greater than the surface tension of the mixture of alcohol and water
d. The surface tension of the olive oil is less than the surface tension of the mixture of alcohol and water

233. A wooden block of volume 0.05 m^3 is floating in water. What is the buoyant force acting on the block? (Density of water = 1000 kg/m^3)
a. 50N b. 500N c. 5000N d. 50000N

234. A fluid is undergoing steady flow. Therefore:
a. the velocity of any given molecule of fluid does not change
b. the pressure does into very from point to point
c. the velocity at any given point does not vary with time
d. the density does not vary from point to point

235. The equation of continuity for fluid flow can be derived from the conservation of:
a. energy b. volume **c. mass** d. pressure

236. Which of the following is NOT a type of fluid dynamics?
a. Aerodynamics b. Hydrodynamics c. Magneto hydrodynamics **d. Electrodynamics**

237. Which of the following is a type of fluid flow?
a. Laminar flow b. Turbulent flow c. Steady flow **d. All of the above**

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238. Which of the following is NOT a factor that affects the viscosity of a fluid?
a. Temperature b. Pressure c. Density **d. Gravity**

239. The coefficient of viscosity is a measure of:
a. The resistance of a fluid to flow b. The force required to move a fluid
c. The velocity of a fluid d. The pressure of a fluid

240. The SI unit of the coefficient of viscosity is:
a. Pascal seconds b. Newton second per meter square c. Poise (PI) **d. All of the above**

241. Which of the following fluids has the highest coefficient of viscosity?
a. Water **b. Honey** c. Air d. Alcohol

242. The coefficient of viscosity of a fluid decreases with:
a. Temperature b. Pressure c. Density d. Molecular structure

243. Which of the following methods can be sued to measure the coefficient of viscosity of a fluid?
a. Capillary viscometer b. Falling ball viscometer
c. Rotational viscometer **d. All of the above**

244. The terminal velocity of a spherical body falling through a fluid is inversely proportional to:
a. The radius of the body b. The density of the body
c. The viscosity of the fluid d. All of the above

245. What is the best way to estimate uncertainty?
a. By taking multiple measurements and calculating the standard deviation
b. By using a variety of instruments
c. By calibrating instruments regularly
d. None of the above

246. What is the purpose of dimensional analysis?
a. To check the consistency of equations b. To derive new equations
c. To estimate the values of physical quantities **d. All of the above**

247. Which of the following measuring instruments is used to measure time:
a. Stopwatch b. Clock c. Pendulum **d. All of the above**

248. Which of the following measuring instruments is used to measure the length of a straight line?
a. Ruler b. Vernier calipers c. Micrometer screw gauge d. All of the above

249. Which of the following is a type of measuring instrument used to measure electrical quantities?
a. Ammeter b. Voltmeter c. Ohmmeter **d. All of the above**

250. Which of the following measuring instruments is used to measure current?
a. Ammeter b. Voltmeter c. Ohmmeter d. All of the above

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251. Which of the following measuring instruments is used to measure voltage?
a. Ammeter **b. Voltmeter** c. Ohmmeter d. All of the above

252. Which of the following measuring instruments is used to measure resistance?
a. Ammeter b. Voltmeter **c. Ohmmeter** d. All of the above

253. Which of the following is a base unit in the SI system?
a. Meter b. Kilogram c. Second **d. All of the above**

254. Which of the following is a derived unit?
a. Velocity b. Force c. Energy **d. All of the above**

255. Which of the following is a supplementary unit?
a. Radian b. Steradian **c. Both a and b** d. None of the above

256. Which branch of physics studies the motion of objects and the forces that act on them?
a. Mechanics b. Thermodynamics c. Electromagnetism d. Quantum mechanics

257. Which branch of physics studies the relationship between heat and other forms of energy?
a. Mechanics **b. Thermodynamics** c. Electromagnetism d. Quantum mechanics

258. Which branch of physics studies the interaction of electric and magnetic fields?
a. Mechanics b. Thermodynamics **c. Electromagnetism** d. Quantum mechanics

259. Which branch of physics studies the behavior of matter and the atomic and subatomic level?
a. Mechanics b. Thermodynamics c. Electromagnetism **d. Quantum mechanics**

260. Which branch of physics studies the behavior of light and other forms of electromagnetic radiation?
a. Optics b. Acoustics c. Nuclear physics d. Particle physics

261. Which branch of physics studies the behavior of sound waves?
a. Optics **b. Acoustics** c. Nuclear physics d. Particle physics

262. Which branch of physics studies the structure and behavior of nuclei?
a. Optics b. Acoustics **c. Nuclear physics** d. Particle physics

263. Which branch of physics studies the behavior of elementary particles, such as electrons, quarks, and photons?
a. Optics b. Acoustics c. Nuclear physics **d. Particle physics**

264. To get a resultant displacement of 10m, two displacement vectors of magnitude 6m and 8m should be combined:
a. Parallel b. Antiparallel
c. At and angle of 45^o **d. Perpendicular to each other**

265. The velocity of a particle at an instant is 10 m/s and after 5 sec the velocity of particle is 20m/s. The velocity 3 sec before in m/s is:
a. 8 **b. 4** c. 6 d. 7

266. A ball is thrown upwards with a velocity of 100 m/s. It will reach the ground after:
a. 10 sec **b. 20 sec** c. 5 sec d. 40 sec

267. Two projectiles are fired from the same point with the same speed at angles of projection 60^o and 30^o respectively. Which one of the following is true?
a. The range will be same b. Their maximum height will be same
c. Their landing velocity will be same d. Their time of flight will be same

268. The ratio of numerical values of average velocity and average speed of a body is always:
a. Unity b. Unity or less c. Unity or more d. Less than unity

269. If the average velocities of a body becomes equal to the instantaneous velocity, body is said to be moving with:
a. Uniform acceleration b. Uniform velocity
c. Variable velocity d. Variable acceleration

270. At the top of a trajectory of a projectile, the acceleration is:
a. maximum b. minimum c. zero **d. g**

271. At what angle the range of projectile becomes equal to the height of projectile?
a. 65^o b. 45^o **c. 76^o** d. 30^o

272. If the dot product of two non-zero vectors vanishes; the vectors will be:
a. in the same direction b. opposite direction to each other
c. perpendicular to each other d. zero

273. If two forces of the same magnitude F are acting at an angle of 120^o with each other the magnitude of their resultant will be:
a. 2F b. 0 c. 0.5 F **d. F**

274. Maximum height of projection depends upon:
a. Angle of projection b. Velocity of projection
c. Both angle and velocity d. Range

275. Which of the following equations represents the relationship between velocity, acceleration, and time in uniformly accelerated motion?
a. $v = u + at$ b. $v = u + at^2$ c. $v = u + 2at$ d. $v = ut + 1/2at^2$

276. If an object moves in a circular path at a constant speed, what can be said about its acceleration?
a. It is directed toward the center of the circle b. It is directed away from the center of the circle
c. It is zero d. It is directed tangentially to the circle

277. If a.b = 0 when a ≠ 0, b ≠ 0, then the vectors are:
a. Parallel   b. Opposite   **c. Perpendicular**   d. Zero

278. When an object is at rest, what is the value of its velocity?
a. Zero   b. Positive   c. Negative   d. Cannot be determined from the information given

279. If a car accelerates uniformly from 10 m/s to 30 m/s in 5 seconds, what is its acceleration?
a. 4 m/s^2   b. 6 m/s^2 c. 8 m/s^2 d. 10 m/s^2

280. If A and B are two vector:
a. A . B = B . A   b. A . B = -B . A   c. A . B = -A . B   d. B . A = - B . A

281. Two bodies launched with different angle from a same point and they had same range. The angle will be:
a. 45^o and 30^o b. 30^o and 70^o **c. 20^o and 70^o**   d. 100^o and 80^o

282. If a projectile is fired at angle of 45^o with the velocity of 100 m/s, it hits the target. It will have double the range if its velocity is:
a. 141.4 m/s   b. 200 m/s   c. 1173.2 m/s   d. 400 m/s.

283. The range of a projectile becomes half of its maximum value if the angle of projectile is:
a. 15^o   b. 45^o c. 60^o d. 90^o

284. Angular velocity is also known as angular:
a. Rotation   b. Speed   **c. Frequency**   d. Acceleration

285. A car is moving with a speed of 20 m/s. It increases its speed to 40 m/s in 5 seconds. What is the magnitude of its acceleration?
a. 12 m/s^2 b. 8 m/s^2 c. 16 m/s^2 **d. 4 m/s^2**

286. A body goes from 2 meter to 8-meter mark and back to 2 meter mark in 3 sec, its average speed is:
a. 2 m/s   b. 6 m/s   **c. 4 m/s**   d. zero

287. Two forces of the same magnitude F make an angle of 180^o with each other their resultant is:
a. zero   b. F   c. $\sqrt{2}$ F   d. 2F

288. The dot product of unit vectors l and k is:
a. zero   b. 1   c. j   d. -1

289. Due to presence of air resistance the total time of flight of a projectile:
a. remains the same   b. decreases   c. becomes zero   **d. increases**

290. If a vector quantity is divided by its magnitude, the vector obtained is called:
a. Unit vector   b. Position vector   c. Null vector   d. Free vector

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291. What does a horizontal line on a velocity time graph represent?
a. constant acceleration **b. zero velocity** c. negative acceleration d. changing velocity

292. On a velocity time graph, the area under the curve represents:
a. displacement b. acceleration c. time d. speed

293. A diagonal line on a velocity time graph indicates:
a. Constant velocity b. Increasing velocity c. Decreasing velocity d. Zero velocity

294. If a velocity time graph is a straight horizontal line, what can be said about the objects motion?
a. The object is at rest b. The object is accelerating
c. The object is moving with constant speed
d. The object is changing direction continuously

295. On a displacement time graph, a curve that is steeper indicates:
a. Slower motion **b. Faster motion** c. Negative displacement d. No motion

296. What does the slope of a displacement time graph represent?
a. Speed b. Acceleration c. Distance **d. Velocity**

297. When $|\vec{A}.\vec{B}| = |\vec{A}+\vec{B}|$ the angle between the vectors $\vec{A}$ and $\vec{B}$ is:
a. Zero b. 45^{o} **c. 90^{o}** d. 180^{o}

298. On a displacement time graph, the area under the curve represents:
a. Speed b. Velocity c. Acceleration **d. Displacement travelled**

299. Two perpendicular vectors having the magnitudes of 4 units and 3 units are added. The resultant of the magnitude is:
a. 7 units b. 12 units c. 25 units **d. 5 units**

300. If $\vec{A}.\vec{B} = \vec{A} \times \vec{B}$, then the angle between A and B is:
a. 90^{o} b. 45^{o} c. 120^{o} d. 180^{o}

301. J x J is equal to:
a. J^2 b. j. **c. zero** d. one

302. The motion of rocket in the space is according to the law of conservation of:
a. Energy **b. Linear momentum** c. Mass d. Angular momentum

303. A bomb of mass 12 kg initially at rest explodes into two places of masses 4kg and 6kg. the speed of 8kg mass is 6 m/s. The kinetic energy the kinetic energy of 4kg mass is:
a. 32 J b. 48 J c. 114 J **d. 288 J**

304. If momentum is increased by 20% then K.E increases by:
a. 44% b. 55% c. 66% d. 77%

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305. The kinetic energy of body of mass 2kg and momentum of 2Ns is:
a. 1 J b. 2 J c. 3 J d. 4 J

306. For the same kinetic energy, the momentum is maximum for:
a. an electron b. a proton c. a deuteron **d. alpha particle**

307. A 3kg bowling ball experience a net force of 15N. What will be its acceleration:
a. 35 m/s^2 b. 7 m/s^2 **c. 5 m/s^2** d. 35 m/s^2

308. Angle of friction and angle of repose are:
a. Equal to each other b. Not equal to each other
c. Proportional to each other d. None of the above

309. A player catches a ball of mass 150gm moving at a rate of 20m/s. If the process of catching is to be completed in 0.1 sec. What is the force exerted by the ball on the hands on the player?
a. 3000 N b. 300 N c. 30 N d. 0.3 N

310. Which of the following is momentum closely related to?
a. Force b. Power **c. Impulse** d. Kinetic energy

311. When a force acts on a ball of mass 150g for 0.1 sec, it produces an acceleration of 20 m/s^2. What is the value of this impulsive force?
a. 0.5 N-s b. 0.1 N-s c. 0.3 N-s d. 1.2 N-s

312. Which is the branch of physics that deals with the motion of a body by considering the cause?
a. Statics b. Thermodynamics **c. Dynamics** d. Astronomy

313. The velocity of a body after elastic collision with a body of the same mass at rest will be:
a. Reduced to half b. Doubled
c. Un-altered **d. Zero**

314. The ratio of limiting friction to the normal reaction is called:
a. Force of friction **b. Co-efficient of friction** c. Kinetic friction d. Static friction

315. In case pf inelastic collision:
a. Both momentum and K.E are conserved b. Neither momentum nor K.E is conserved
c. Only momentum is conserved d. Only K.E is conserved

316. When a body slides over a surface the kinetic friction (f_k) and the static friction (f_s) are related as:
a. $f_k < f_s$ b. $f_a < f_k$ c. $f_k = 0$ d. $f_a = 0$

317. When a constant force is applied on a body it moves with:
a. Constant speed b. Constant velocity **c. Constant acceleration** d. Constant displacement

318. The rate of change of linear momentum is equal to:
a. Acceleration b. Torque **c. Force** d. Angular velocity

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319. When a body is placed at an inclined frictionless plane the force by which the body slides down:
a. mgcos0 **b. mgsin0** c. gcos0 d. zero

320. The ratio of kinetic friction plane at which a body just begins to slide down the plane is called:
a. Angle of repose b. Angle of Elevation c. Angle of Projection d. Angle of Incidence

321. A body of mass 'm' moving with velocity 'V' collides with a stationary body of mass '2m' and combine to form one body. The velocity of the combine body after collision, is:
a. 3V **b. V/3** c. 9V d. 4V

322. The rotational inertia of a wheel about its axle does not depend upon its:
a. Diameter b. Distribution of mass c. mass **d. speed of rotation**

323. A force with at given magnitude is to be applied to a wheel. The torque can be maximized by:
a. applying the force near the axle, radially outward from the axle
b. applying the force near the rim, radially outward
c. applying the force near the axle, parallel to a tangent to the wheel
d. applying the force at the rim, tangent to the rim

324. An object rotating about a fixed axis, I is its rotational inertia and alpha is its angular acceleration. Its:
a. is the definition of torque b. is the definition of rotational inertia
c. is definition of angular acceleration **d. follows directly from Newton's second law**

325. The angular momentum vector of Earth about its rotation axis, due to its daily rotation is, directed:
a. tangent to the equator towards east b. tangent to the equator towards the west
c. north d. towards the sun

326. A stone of 2kg is tied to a 0.50m long string and swung around a circle at constant angular velocity of 12 rad/s. The net torque on the stone about the center of the circle is:
a. 0 N.m b. 12 N.m c. 6 N.m d. 72 N.m

327. A man, with his arms at his sides, is spinning on a light frictionless turntable. When he extends his arms:
a. his angular velocity increases b. his angular velocity remains same
c. his rotational inertia decreases d. his angular momentum remains the same

328. A space station revolves around the earth as a satellite, 100km above the Earth's surface. What is the net force on an astronaut at rest inside the space station?
a. equal to her weight on earth b. a little less than her weight on earth
c. less than half her weight on earth **d. zero (she is weightless)**

329. If the external torque acting on a body is zero, then its:
a. angular momentum is zero **b. angular momentum is observed**
c. angular acceleration is maximum d. rotational motion is maximum

330. A body moving along a circular path may have constant
a. Velocity **b. Acceleration** c. Momentum d. Speed

331. The angle between centripetal acceleration and tangential acceleration is:
a. 0^o b. 45^o c. 30^o **d. 90^o**

332. If 'r' is the radius of the circular path of particle, the relation between its linear and angular velocities are:
a. v = r x w b. v = µ x w c. w = v x r d. w = r x v

333. One radian is equal to:
a. 57.3^o b. 0.017^o c. 67.3^o d. 180^o

334. The dimension of angular velocity are:
a. T^{-1} b. LT^{-1} c. MLT^{-1} d. LT^{-2}

335. A body is moving along a circle with increasing speed it possess:
a. Tangential acceleration only
b. Centripetal acceleration only
c. No acceleration
d. Both tangential and centripetal acceleration

336. Every point on the rotating body has the same:
a. Linear velocity **b. Angular velocity** c. Angular momentum d. Torque

337. When a body moves with a constant speed in a circle:
a. Its velocity is changing
b. Its acceleration is zero
c. Its acceleration is increasing
d. Its velocity is uniform

338. Centripetal force is also called:
a. Centre-seeking force
b. Centrifugal force
c. Tangential force
d. Gravitational force

339. When a body moves along circumference of a circle with uniform speed. Changes takes place in its:
a. Linear velocity
b. Tangential acceleration
c. Radius of circle
d. Linear velocity and tangential acceleration

340. Angular velocity is also known as angular:
a. Rotation b. Speed **c. Frequency** d. Acceleration

341. A stone weighting 1kg is tied with 1m long rope and whirled on a circular path with linear speed 1 m/s, tension in the string will be:
a. 9.8N . b. 3.14N **c. 1N** d. 100N

342. A car is moving on a banked curve at a constant speed. Which of the following forces is the centripetal force?
a. The weight of the car
b. The normal reaction of the road on the car
c. The frictional force between the tires of the car and the road
d. The sum of the normal reaction and the frictional force

343. A satellite is orbiting the Earth in a circular orbit. IF the orbital radius if the satellite is douvled, what will happen to its orbital velocity?
a. $2MR^2/5$ b. $5MR^2/2$ c. $7MR^2/5$ d. $9MR^2/5$

344. Which of the following laws states that the sum of the torques on a body is equal to the moment of inertia of the body times its angular acceleration?
a. Rotational equivalent of Newton's second law
b. Newton's second law of motion
c. Law of conservation of angular momentum
d. Law of conservation of linear momentum

345. What is the direction of torque?
a. The direction of the torque is perpendicular to the plane of the force and distance
b. The direction of the torque is the same as the direction of the force
c. The direction of the torque is the opposite of the direction of the force
d. The direction of the torque is the direction of the distance

346. What is the law of conservation of angular momentum?
a. The angular momentum remains constant if there is no net torque acting on system
b. The angular momentum is proportional to the net torque acting on the system
c. The angular momentum is inversely proportional to the net torque acting on system
d. The angular momentum is independent of the net torque acting on the system

347. Work done by centripetal force is always:
a. Maximum b. Minimum **c. Zero** d. None of these

348. A body of mass 5kg is moving with a momentum of 10kg m/s. A force of 0.2N acts on it in the direction of motion of the body for 10 seconds. The increase in kinetic energy is:
a. 2.8 J b. 3.2 J c. 3.8 J d. 4.4 J

349. The kinetic energy of a light and a heavy object is same. Which object has maximum momentum?
a. Light object b. Heavy object c. Both have same momentum d. N.O.T.

350. Two bodies of mass 1kg and 2kg have equal momentum. Then the ratio of their kinetic energies is:
a. 2 : 1 b. 3 : 1 **c. 1 : 3** d. 1 : 1

351. A body fallen from height h. After it has fallen a height h/2, its will possess:
a. only potential energy b. only kinetic energy
c. half potential half kinetic energy d. more kinetic and less potential

352. Which of the following quantity can be calculated by multiplying force and velocity?
a. acceleration **b. power** c. torque d. work

353. The minimum velocity given to an object so that it emerges out from the gravitational field of earth is about:
a. 11.2km/s b. 15.3 km/s c. 5 km/s d. 9.8 km/s

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354. When one joule of work is done on a body in one second, power of body is said to be:
    a. One watt b. 0.5 watt c. zero d. 100 watt

355. The absolute potential energy of an object depends on:
    a. The object's mass and height b. The object's mass and speed
    c. The object's shape and size d. The object's color and temperature

356. The escape velocity of a planet depends on which of the following factors?
    a. The mass of the planet only b. The radius of the planet only
    c. Both the mass and the radius of the planet d. The density of the planet

357. The work done by the conservative force along a closed path is:
    a. Positive b. Negative **c. Zero** d. None of these

358. The dot product of force and velocity is:
    a. Work b. Power c. Energy **d. Momentum**

359. In case of negative work, the angle between work the angle between the force and displacement is:
    a. zero b. 90^o c. 180^o d. 45^o

360. If mass and speed both are double, then the kinetic energy of a moving body:
    a. Increase 4 times b. Increase 6 times
    c. Increase 8 times d. Increase 10 times

361. Kilo-watt hour is the unit of:
    a. Energy b. Power c. Time d. Velocity

362. One horse power is equal to:
    a. 400 watt b. 550 watt **c. 746 watt** d. 1055 watt

363. If F = 4i – 2j and d = 3i + 4j the work done will be:
    a. 4 Joules b. 8 Joules c. 12 Joules d. 2 Joules

364. Watt hour is the unit of:
    a. Energy b. Power c. Time d. Velocity

365. When the direction of force is opposite to the direction of displacement, than the work done will be:
    a. Positive **b. Negative** c. Zero d. None of these

366. Which of the following is not the unit of power:
    a. Kilowatt hour b. Horse power c. Foot-ibs/sec d. Joules/sec

367. The work done in moving a body between two points in a conservative field is independent of:
    a. Direction b. Force applied **c. Path followed by a body** d. Power

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368. The dimension of power are:
a. ML^2T^{-3} b. $M^2L^2T^{-3}$ c. $M^2L^2T^{-1}$ d. ML^2T^2

369. The dimension of K.E are:
a. ½ ML^2T^{-3} b. ½ $M^2L^2T^{-3}$ c. $ML^{-1}T^{-2}$ **d. ML^2T^{-2}**

370. Potential energy is increased when the work is done:
a. Along the field **b. Against the field** c. By the field d. All of these

371. The work done in lifting 30kg of brick to a height of 20m is:
a. 2800J b. 600J **c. 5880J** d. 2910J

372. Which of the following force is not the conservative:
a. Coulombs force b. Force of a compressed elastic spring
c. Frictional force d. Gravitational force between two masses

373. A completely submerged object always displaces its own:
a. weight of fluid **b. density of a fluid** c. volume of a fluid d. Area of fluid

374. The pressure exerted on the ground by a man is greatest when:
a. he stands with both feet flat on the ground b. the ground
c. he stands on the toes of one foot d. he lies down on the ground

375. In a stationary homogenous liquid:
a. pressure is the same at all points b. pressure depends on the direction
c. pressure is intendent of any atmospheric pressure on the upper surface of the liquid
d. pressure is the same at all points at the same level

376. One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Ap the pressure at the larger piston:
a. Increases by 2Δp b. Increases by Δp c. Increases by Δp/2 d. Increases by 4Δp

377. In a vacuum, an object has:
a. No buoyant force b. no weight c. no mass d. none of these

378. The pressure at the bottom of a pond does NOT depend on:
a. Water density b. the surface area of the pond
c. the depth of the pond d. none of these

379. A rock suspended by a weighting scale weighs 5N out of water and 3N when submerged in water. What is the buoyant force on the rock?
a. 3N b. 8N c. 5Nq **d. 2N**

380. "An object completely submerged in a fluid displaces its own volume of fluid. This is:
a. Pascal's paradox **b. Pascal's principle** c. Archimedes' principle d. true, but none of the above

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381. Salt water has greater density than freshwater. A boat floats in both freshwater and saltwater. The buoyant force on the boat in salt water is that in freshwater.
a. equal to b. larger than **c. smaller than** d. same as

382. You fill a tall glass with ice and then add water to the level of the glass's rim, so some fraction of the ice floats above rim. When the ice melts, what's happens to the water level?
a. the water overflows the rim
b. the water level drops below the rim
c. the water level stays at the rim
d. it depends on the difference in density between water and ice

383. Which of the following is NOT a fluid?
a. Water b. Air **c. Sand** d. Oil

384. Which of the following is a property of fluids?
a. They can flow
b. They can take the shape of their container
c. They are incompressible
d. All of the above

385. What is the difference between a liquid and a gas?
a. Liquids have a definite shape, while gases do not
b. Liquids are less compressible than gases
c. Liquids have a higher viscosity than gases
d. All of the above

386. What is the SI unit of pressure?
a. Pascal (Pa) b. Newton (N) c. Kilogram (kg) d. Meter (m)

387. What is the difference between a compressible and an incompressible fluid?
a. A compressible fluid can change its volume, an incompressible fluid cannot
b. A compressible fluid has a higher density than an incompressible fluid
c. A compressible fluid has a higher viscosity than an incompressible fluid
d. All of the above

388. Pascal's law states that:
a. Pressure is inversely proportional to the area
b. Pressure is directly proportional to the volume
c. Pressure is transmitted equally in all directions through a fluid at rest
d. Pressure is directly proportional to the temperature

389. Which of the following is NOT an example of Pascal's law in action?
a. A hydraulic press b. A car jack c. A syringe **d. A boat floating on water**

390. Which of the following is an application of Pascal's law?
a. A hydraulic brake system
b. A submarine
c. A water balloon
d. All of the above

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391. Which of the following factors will affect the buoyant force acting on an object?
a. The volume of the object
b. The density of the fluid
c. The gravitational acceleration
d. All of the above

392. A pipe has a diameter of 10cm and a flow rate of 10 liters per second. What is the velocity of the fluid in the pipe?
a. 2 m/s
b. 20 m/s
c. 10 m/s
d. 1.27 m/s

393. According to the law of floatation, an object will float if:
a. The weight of the object is greater than the buoyant force acting on it
b. The weight of the object is equal to the buoyant force acting on it
c. The weight of the object is less than the buoyant force acting on it
d. The weight of object becomes infinite

394. Which of the following factors will affect the buoyant force acting on an object?
a. The volume of the object
b. The density of the fluid
c. The gravitational acceleration
d. All of the above

395. Surface tension is defined as:
a. The force per unit length acting on a fluid surface
b. The pressure difference between the inside and outside of a liquid drop
c. The force required to break a liquid surface
d. All of the above

396. The SI unit of surface tension is:
a. Newton per meter (Nm^{-1})
b. Pascal (Pa)
c. Joule (J)
d. None of the above

397. Which of the following liquids will have the highest capillary rise in a glass tube?
a. Water
b. Mercury
c. Ethanol
d. Soap solution

398. A 2μC point charge is located a distance 'd' away from 6μC point charge, what is the ratio of F_{12}/F_{21}?
a. 1/3
b. 3
c. 1
d. 12

399. The minimum charge on an object cannot be less than:
a. 1.6×10^{-19}C
b. 3.2×10^{-19}C
c. 9.1×10^{8}
d. No definite value exist

400. Two charges are placed at a certain distance. IF the magnitude of each charge is doubled the force will become:
a. 1/4th of its original value
b. 4 times of its original value
c. 1/8th of its original value
d. 8 times of its original value

401. Which of the following can be deflected while moving in the electric field?
a. neutron
b. photon
c. electron
d. both a and b

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402. The flux through a flat surface of are 'A' in a uniform electric field 'E' is maximum when the surface area is:
a. Parallel to E **b. perpendicular to E** c. placed 45^o to E d. placed 60^o to E

403. The product of charge 'q' and small separation 'd' between two charges of same magnitude and opposite in nature is known as:
a. Electric dipole b. Moment arm **c. Electric dipole moment** d. Flux of electric field

404. 12J of work is to be done against an existence electric field to take a charge of 0.01 C from one-point A to another point B. The potential difference between B and A is:
a. 120 V **b. 1200 V** c. 1.2 V d. 12 V

405. The force between two charges placed in air is F. If air is replaced by a medium of relative permittivity ε_r then force is reduced to:
a. F ε_r b. F / ε_r c. ε_r / F d. ε_0 ε_r

406. The negative gradient of the potential is:
a. Potential energy b. Voltage **c. Electric field intensity** d. Electric flux

407. The electric flux through a plane area will be half of its maximum value when area is held at angle of with electric field:
a. 30^o **b. 60^o** c. 45^o d. 90^o